

Handland Mark Detection Using OpenCV

Group Number: 1

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Abstract

- This paper introduces a straightforward yet powerful approach for hand landmark detection using the combination of OpenCV, Mediapipe, and Python.
- Leveraging the capabilities of Mediapipe's pre-trained models for hand landmark detection and the versatile image processing tools offered by OpenCV, our method achieves accurate and real-time detection of landmarks on human hands.
- By integrating these libraries into a Python-based workflow, we demonstrate the ease of implementation and effectiveness of our approach.

Background Information

OpenCV:

1. Open-source computer vision library.
2. Offers image processing, object detection, and video analysis functionalities.

MediaPipe:

1. Developed by Google for machine learning-based perception tasks.
2. Provides pre-trained models and APIs for tasks like hand tracking and pose estimation.

Python Language:

1. High-level programming language used in data science, machine learning, and web development.
2. Essential for developing applications using OpenCV, MediaPipe, and other libraries.

Literature Survey-1

SL.NO.	AUTHORS	METHODS	MERITS	DEMERITS
1	Hand Landmark-Based Sign Language Recognition Using Deep Learning Jerry John, Bismin V Sherif. Machine Learning and Autonomous Systems: Proceedings of ICMLAS 2021, 147-157, 2022	●Acquiring a dataset containing images of hand gestures representing sign language alphabets. ●Preparing the data by cleaning and normalizing images. Extracting hand skeletons from images to create a skeleton dataset for more precise feature extraction.	●The user-independent model allows for broader applicability, accommodating individuals with different hand shapes and sizes. ●Enabling speech-impaired individuals to communicate with those unfamiliar with sign language by translating hand gestures into English sentences.	●Acknowledging the difficulty of accurately classifying certain sign language symbols that look alike, which may affect the overall performance of the image-based classification model. ●Deep learning models, especially those involving neural networks, can be complex. This complexity might lead to challenges in training, deployment, or resource requirements.

Literature Survey-2

SL.NO.	AUTHORS	METHODS	MERITS	DEMERITS
2	VIRTUAL MOUSE WITH HAND GESTURE RECOGNITION BASED ON HAND LAND-MARK MODEL FOR POINTING DEVICE Jeffri Dian Asmoro, Achmad Teguh Wibowo, Mujib Ridwan. JURTEKSI (Jurnal Teknologi dan Sistem Informasi) 9 (2), 261-268, 2023	●Implement machine learning techniques to recognize and interpret hand gestures captured by a webcam. ●This involves training the system to associate specific gestures with corresponding commands. Integrate webcam functionality to capture and process hand gestures, allowing users to control the virtual mouse through hand movements.	●Suitable for desktop devices, making it adaptable for a wide range of users and applications. ●Uses a common and widely available hardware component, the webcam, making it easily implementable on devices with built-in or external cameras.	●Depending on the complexity of the hand landmark model and machine learning algorithms, there might be limitations on the variety of gestures that can be reliably recognized. ●The system relies on the availability and functionality of a webcam, potentially limiting its use in environments without camera-equipped devices.

Literature Survey-3

SL.NO.	AUTHORS	METHODS	MERITS	DEMERITS
3	Digital Handwriting Recognition Using Hand Tracking by Using Media Pipe and OPENCV Libraries Mr M Nagaraju, MD Sohail, B Rahul Teja, NS Siddharth	●Utilize the MediaPipe library, specifically its hand tracking capabilities, to recognize and track hand movements in the video stream. ●Extract landmark positions (0-20) representing various points on the hand. Map the detected hand landmarks to a digital representation of handwriting.	●The application is easily accessible as it only requires a webcam, making it a practical solution for individuals who do not have specialized writing tablets. ●Provides a real-time interactive experience for users, allowing them to write, erase, and modify content seamlessly.	●Acknowledges that the digital handwriting experience may not fully match the natural feel of writing on paper or using specialized writing tablets. ●The accuracy of hand tracking and landmark detection can be influenced by lighting conditions and the quality of the webcam.

Proposed System

Feature 1: Real-time hand landmark detection

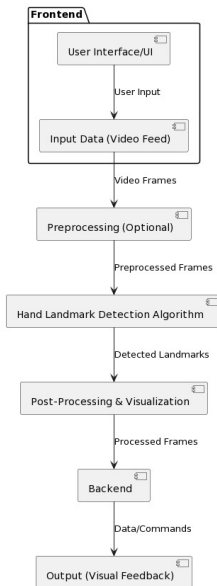
Feature 2: Visualization of detected landmarks on the hand

Feature 3: Ability to track hand movements

Feature 4: Integration with other applications for gesture recognition

Feature 5: Exporting detected landmarks data for analysis

System Design



Technology Used

1 . OpenCV (Open Source Computer Vision Library):

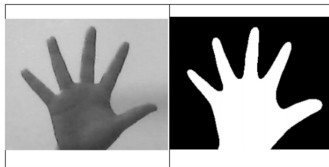
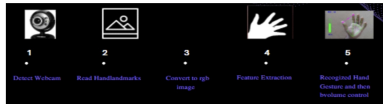
OpenCV is a popular open-source library for computer vision and image processing tasks.

2. Python Programming Language:

Python is often the language of choice for implementing computer vision applications using OpenCV due to its simplicity, readability, and extensive library support.

3. Hardware:

A webcam or other image capture device is used to acquire video frames for hand landmark detection in real-time applications



Result And Discussion

- Results indicate high accuracy in landmark detection with reasonable performance under various conditions.
- The system performs well in diverse scenarios but may struggle with extremely fast movements or poor lighting.
- Future improvements could include refining the detection algorithms and exploring applications in more complex augmented reality systems.

Result And Discussion

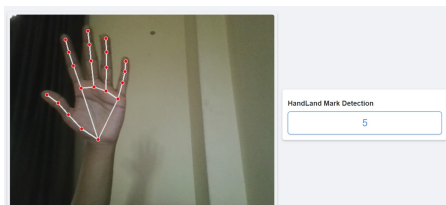


Figure: Final Output

Conclusion and Future Scope

CONCLUSION

- 1. Successful Implementation:** The hand landmark detection project has effectively implemented a real-time system capable of accurately tracking landmarks on human hands.
- 2. Integration of Technologies:** By leveraging machine learning models, computer vision techniques, and web technologies, the system provides users with an intuitive interface for interaction.
- 3. Efficient Processing Pipeline:** Integration with OpenCV ensures efficient preprocessing, feature extraction, and visualization tasks within the detection pipeline.
- 4. Smooth Communication:** Real-time communication between frontend and backend facilitates smooth streaming of webcam feeds and prompt delivery of detected landmarks.

Conclusion and Future Scope

FUTURE SCOPE

- 1. Accuracy Enhancement:** Future improvements aim to fine-tune machine learning models and explore advanced computer vision techniques to enhance detection accuracy.
- 2. Gesture Recognition Expansion:** Extending the system to recognize specific hand gestures or sign language will broaden its applicability and usefulness.
- 3. Multi-modal Integration:** Integration with additional modalities such as voice recognition or depth sensing will further enrich the interaction experience.
- 4. Mobile Deployment:** Adapting the system for deployment on mobile devices will enable hand landmark detection on-the-go, increasing its accessibility and utility.

References

- [1] John, J., Sherif, B. V. (2022). Hand Landmark-Based Sign Language Recognition Using Deep Learning. In Machine Learning and Autonomous Systems: Proceedings of ICMLAS 2021 (pp. 147-157).
- [2] Asmoro, J. D., Wibowo, A. T., Ridwan, M. (2023). Virtual Mouse with Hand Gesture Recognition Based on Hand Landmark Model for Pointing Device. JURTEKSI (Jurnal Teknologi dan Sistem Informasi), 9(2), 261-268.
- [3] Nagaraju, M., Sohail, M. D., Teja, B. Rahul, Sidhharth, N. S. (2022). Digital Handwriting Recognition Using Hand Tracking by Using Media Pipe and OpenCV Libraries, 2022
- [4] K. S. Varun, I. Puneeth and T. P. Jacob, "Virtual Mouse Implementation using Open CV," 2019 3rd International Conference on Trends in Electronics and Informatics (ICOEI), Tirunelveli, India, 2019, pp. 435-438, doi: 10.1109/ICOEI.2019.8862764.